

Sahana Aster KŌWHAI, Ph.D.

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Associate Professor of Palaeoclimatology, Te Tari Tātai Arowhenua — Department of Geology
University of Ōtākou — Te Whare Wānanga o Ōtākou,
Southern Ocean Research Centre, Te Waka Wānanga, Ōtepoti Dunedin, Aotearoa New Zealand

SHORT BIO

Sahana Aster Kōwhai is a **fictional** palaeoclimatologist whose *research* reconstructs Southern Ocean change from marine sediment archives.

RESEARCH FOCUS

Southern Ocean palaeoceanography across glacial terminations, with an emphasis on multiproxy reconstructions, reproducible chronology and equitable Antarctic field partnerships.

ACADEMIC APPOINTMENTS / EXPERIENCE

- Associate Professor of Palaeoclimatology** Feb 2021 – Present
University of Ōtākou — Te Whare Wānanga o Ōtākou, Department of Geology Ōtepoti Dunedin, Aotearoa New Zealand
 - Cross-appointed affiliate of the Southern Ocean Research Centre, Te Waka Wānanga.
- Senior Lecturer in Marine Geology** Jul 2016 – Jan 2021
University of Southern Tasmania, Institute for Marine and Southern Ocean Studies nipaluna Hobart, Australia

EDUCATION

- Ph.D. in Marine Geology** 2010 – 2014
Commonwealth University of Australia Canberra, Australia
 - Thesis: Foraminiferal Mg/Ca and Southern Ocean warming across Termination I.
- M.Sc. in Earth Sciences (First Class Honours)** 2008 – 2009
Cape Maritime University Cape Town, South Africa
- B.Sc. in Geology and Oceanography** 2005 – 2008
Madras Institute of Ocean Sciences Chennai, India

TEACHING

- Course coordinator and lecturer** 2021 – Present
University of Ōtākou — Te Whare Wānanga o Ōtākou Ōtepoti Dunedin, Aotearoa New Zealand

Course	Programme	Topics	Points
GEOL 273	B.Sc. Geology	Palaeoclimate archives; proxy calibration	18
GEOL 472	B.Sc. (Hons)	Marine cores; age-depth modelling	18
OCEN 405	M.Sc. Oceanography	Southern Ocean circulation; uncertainty	20

EDITORIAL, REVIEWING & CONFERENCE SERVICE

- Associate Editor** 2023 – Present
Palaeoceanography and Climate Systems, Marine palaeoclimate
- Associate Editor**
Example Journal of Palaeoclimate Studies, 2022–2024 [IF: 3.4, Q1]
- Scientific Steering Committee**
Southern Ocean Climate Observatory Consortium, 2022–2026
- Panel reviewer** 2021 – Present
Aotearoa Research Council — Te Kāhui Pūtaiao
- Reviewer**
Quaternary Palaeoenvironment Reviews

RESEARCH PROJECTS

- **TIDE — Termination Ice Dynamics and Ecosystems**

Aotearoa Research Council — Te Kāhui Pūtaiao, Principal investigator

◦ Multiproxy reconstruction of Southern Ocean fronts across Termination II, with voyage support from the [Southern Archive Voyage Consortium](#).
- 2024 – 2027
- **Open Core Chronologies**

Austral Polar Science Partnership, Co-investigator

◦ Reusable age-depth models and provenance for legacy sediment cores.
- 2022 – 2025

AWARDS & SCHOLARSHIPS

- **Early Career Research Excellence Award**

Geoscientific Society of Aotearoa

2019
- **ICPR Visiting Fellowship**

International Council for Polar Research

Cambridge, United Kingdom

2017

STUDENT SUPERVISION (THESES & INTERNSHIPS)

Current supervision: projects span marine geochemistry, chronology and open polar data.

Level	#	Notes (scope and international experience)
Doctoral candidates	4	Primary supervisor for two; co-supervisor for two
M.Sc. and honours researchers	9	Completed and current research dissertations

ADDITIONAL INFORMATION

Languages: Tamil (Native), English (Fluent), te reo Māori (Intermediate learner)

PUBLIC ENGAGEMENT

- **Sediment stories: reading climate from ocean cores**

Kōwhai Coast Learning Collective, Community workshop for secondary-school science teachers

Ōamaru, Aotearoa New Zealand

Sep 2025
- **Core archive open day**

Aotearoa Marine Memory Trust, Public demonstration of core scanning and age-model uncertainty

Porirua, Aotearoa New Zealand

Mar 2024
- **Antarctic archives teacher residency**

Polar Evidence Studio, Short course on proxy evidence for secondary-school teachers

Ōtepoti Dunedin, Aotearoa New Zealand

Jul 2023

TALKS & PRESENTATIONS

I=INVITED, C=CONFERENCE

Invited and keynote talks

- [I1] S. A. Kōwhai. “What an open sediment record should remember”. World Congress on Palaeoceanography (Bergen, Norway). Invited keynote. June 18, 2026.

Conference presentations

- [C1] S. A. Kōwhai and M. Te Rangi. “First results from the Pukaki Rise coring transect”. Aotearoa Antarctic Science Symposium (Te Whanganui-a-Tara Wellington, Aotearoa New Zealand). Oral presentation. Aug. 21, 2025.
- [C2] S. A. Kōwhai. “Age-depth ensembles without hidden decisions”. Australasian Quaternary Research Society Biennial Meeting (nipaluna Hobart, Australia). Poster presentation. Dec. 4, 2024. URL: <https://talks.example.edu/kowhai2024ensembles>.
- [C3] S. A. Kōwhai. “From core description to an auditable climate record”. Southern Archives Methods Forum (Ōtepoti Dunedin, Aotearoa New Zealand). Sept. 14, 2023.

PUBLICATIONS

J=JOURNAL, C=CONFERENCE, B=BOOKS, D=DATA, R=UNDER REVIEW

Journal articles (peer-reviewed)

- [J1] S. A. Kōwhai, M. Te Rangi, and C. Okafor. “Southern Ocean front migration recorded across two glacial terminations”. In: *Palaeo-Environments, Palaeoclimates, Palaeoecology* 612 (2026), p. 111001.
- [J2] S. A. Kōwhai, H. Williams, and T. Ndlovu. “Transparent age-depth ensembles for Southern Ocean sediment archives”. In: *Earth’s Past Climate* 20.8 (2024), pp. 1701–1724. URL: <https://publications.example.edu/kowhai2024chronology>.

Conference papers (peer-reviewed)

- [C1] S. A. Kōwhai and M. Te Rangi. “Automated tephra isochron matching in Southern Ocean sediment cores”. In: *Proceedings of the International Palaeoceanography Symposium*. Bergen, Norway, 2023, pp. 145–152. DOI: [10.5555/palaeo.2023.0145](https://doi.org/10.5555/palaeo.2023.0145).

Books and chapters

- [B1] S. A. Kōwhai. "Proxy agreement and disagreement in high-latitude marine archives". In: *Southern Ocean Palaeoceanography: Methods and Records*. Ed. by L. Navarro and K. Adeyemi. Cape Town: Aster Academic Press, 2023, pp. 81–108.
- [B2] S. A. Kōwhai and C. Okafor. "Southern Ocean Foraminiferal Assemblages as Thermal Proxies". In: *Advances in Marine Palaeoecology*. Ed. by L. Navarro. Aster Academic Press, 2022. Chap. 4, pp. 89–112.
- [B3] S. A. Kōwhai and M. Te Rangi. *Field Notes for Reciprocal Polar Science*. Ōtepoti Dunedin: Koru Open Books, 2021.

Data archives

- [D1] S. A. Kōwhai and C. Okafor. *Pukaki Rise multiproxy core chronology, version 1.0*. Aotearoa Polar Data Commons, 2025. URL: <https://data.example.edu/pukaki-rise-v1>.

Manuscripts under review

- [R1] S. A. Kōwhai and M. Te Rangi. "Shelf-water change at the Pacific Antarctic margin". In: *Quaternary Palaeoenvironment Reviews* (2028).
- [R2] S. A. Kōwhai and T. Ndlovu. "Sea-ice retreat and productivity at the East Antarctic margin". In: *The Polar Cryosphere* (2027).

Example record. This curriculum vitae describes a fictional scholar and is supplied only to demonstrate the template.